Recorded processing covariates correct a track on the pairs they were trained on, but the correction does not carry to unseen pairs

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CANDI project, private close-out draft, November 3, 2026.

Abstract

We asked whether recorded processing covariates carry enough information to map one measurement of an epigenomic track onto another of the same material. We built counterfactual arms of 7 tracks, each changing one of 10 processing settings, and trained networks that read the covariates and output the correction. On trained depth pairs the better designs close 0.77 to 0.92 of the gap from a covariate-free map to the oracle (seed range at most 0.53 of the gap). On unseen pairs the gain over a scrambled-covariate twin falls from 0.29 to 0.11, and no design follows the depth law; three final experiments did not change this, but a mappability input helps in low-mappability bins. All readings are exploratory, on 7 tracks.

Keywords: epigenomics, experimental covariates, counterfactual data, sequencing depth

1 Motivation and background

In the ENCODE Imputation Challenge, an average-activity baseline beat all but two of 23 submitted models [1]. The cause was a small processing difference: the single-end training data and the paired-end blind test data were deduplicated by different rules. Processing settings (covariates) thus leave artifacts that a model must correct.

CANDI imputes tracks from raw counts and reads covariates (depth, run type, read length) on both sides. Inside CANDI this conditioning cannot be tested: its corpus pairs an experiment only with thinned copies of itself, so only depth ever changes with the material fixed, and the covariates are collinear (they change together). We therefore built an isolated testbed. Our question: do the recorded covariates carry enough information to map one measurement of a track onto another of the same material?

2 Methods

Counterfactual arms.

We took 7 challenge tracks (one DNase-seq, three narrow and three broad histone marks), processed each to a base of 30M reads (DNase 50M) and to arms that change one setting: depth (15M, 7.5M, 3.75M), read length (36, 50 bp), mapping filter, duplicate removal, paired-end, an antibody proxy (50% or 90% control reads), fragment extension and three control settings: 130 products, each split into two random halves of its reads.

Model and competitors.

A generator g reads source and target covariates C, C' and outputs a map f; $f(X)$ predicts the target as negative binomial (NB) counts per 25 bp bin (Fig. 1a). Four designs of f rise in capacity: per-bin affine (A), per-bin monotone

curve (B), kernel plus curve (C) and a conditioned convolutional network (D). Competitors are a scrambled-covariate twin, one-warp (a covariate-free monotone map per pair) and an oracle (exact binomial thinning for depth). The score is the NB CRPS (continuous ranked probability score, distance from forecast to observation), always split into a shape part and a scale error. A gain counts when it exceeds 2× the seed range (largest minus smallest over 3 seeds). Scoring uses held-out chr19 and chr21, on trained pairs and on unseen pairs (arm to arm, never paired in training). The depth law says a depth change scales the predicted total by the depth ratio.

Final experiments.

Three claims were pre-registered (fixed in writing before any run) on one model shared by all tracks (Fig. 1b): adding a cross-assay imputation or masked-bin denoising objective (weights 0.25, 1, 4) changes unseen-pair CRPS; routing through the base (arm → base → arm′) beats the direct route in two thirds of unseen cells (6 seeds); a 36-mer mappability input lowers CRPS in the lowest-mappability tenth of bins on read-length and mapping-filter arms, beside a 5 Mb shifted-mappability control. After a one-seed smoke test (a short trial run), and before the full runs, the design was amended: scrambled twins were added, a collapsed consistency loss was replaced by an agreement loss between routes, and empty top strata were redefined. The joint probability of meeting every check was 41% to 51%.

3 Results and discussion

Which settings act, and where.

Whole-read resampling sets the floor: at 3.75M reads per half the narrow 1 kb shape floor is 0.79, not 0.69 (Fig. 2b). Peak height moves beyond this floor in 27 of 27 setting rows, shape in only 3 of 27. Signal level explains 0.18 to 0.38 of the per-bin disagreement; regional features add at most 0.0003. Depth acts uniformly (residual 0.774 against its null 0.773); the mapping filter and read length act in low-mappability bins (feature gain 0.099 to 0.110 and 0.013 to 0.042; Fig. 2c).

Trained pairs: the covariates carry the information.

From base to arm, designs B to D close 0.77 to 0.92 of the gap in every mark class (Fig. 2a; seed range at most 0.53 of the gap), above 2× the seed range in 18 of 18 cells. In narrow marks, design B scores 0.175 (0.174 shape + 0.0008 scale error), one-warp 0.197 (0.192 + 0.0057) and the oracle 0.169 (0.169 + 0.0000); 2× the seed range is 0.0015. The affine design A shows no gain (−0.62 to −0.01): a monotone curve gives the main step, and more capacity adds little. For paired-end, the model beats one-warp by 0.055 to 0.213 in 16 of 16 histone cells (2× seed range 0.002 to 0.065). A depth offset helps only A, in scale error (DNase 3.06 ± 0.113 to 1.247 ± 0.016; Table 1). From predicted counts, the peak caller's formula plus a learned correction gives p best (log CRPS 0.0659 ± 0.0035, 0.0656 + 0.0003).

Unseen pairs: the correction does not transfer.

The relative count gain over the twin falls from 0.29 on trained pairs to 0.11 on unseen pairs. The real arms obey the depth law within 0.71%, yet every design meets it on trained pairs (0.05 to 0.12, median $|\log_2|$ error) and misses it on unseen pairs (0.24 to 0.63; Fig. 3a). Removing the pseudocount, a Poisson mean loss, wider networks and an encoder-decoder do not close the miss. Chaining two trained maps through the base does (0.09 to 0.11). The miss is a failure to transfer, not a lack of capacity.

Arm-to-arm training teaches scale, not distribution.

One added arm-to-arm pair per track closes 0.674 of the depth-law gap to the chained map (bar 0.5), but CRPS gets worse: 0.573 (0.466 + 0.107, seed range 0.062) against 0.525 (0.473 + 0.052, 0.032) direct and 0.435 (0.429 + 0.006, 0.003) chained, a gap closed of −0.53. Training on every pair reaches the chained map (0.428, 0.425 + 0.003, 0.005; not pre-registered).

Trained CANDI.

Told a 36 bp read length, CANDI moves its predicted level by −0.89 to −0.22 $\log_2$ (real −1.08 to −1.05); told paired-end, by −0.98 to −0.19 in histone marks (real −0.08 to 0.01) and 0.52 to 0.70

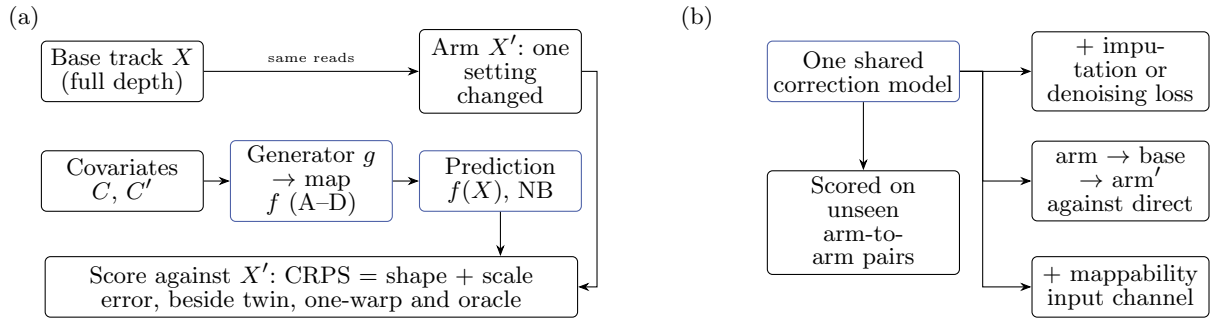


Fig. 1 Design. (a) Each arm reprocesses the base's reads with one setting changed; the generator reads both covariate sets and outputs the map that turns the base into a predicted arm. (b) The three final experiments, each a change to one shared model.

in DNase (real −0.03; Fig. 3b). Its depth control failed, so per-region tracking cannot be read. The PI's reading: CANDI uses both covariates, but not in the direction the real tracks move.

Joint objective: no change.

At weight 1, the cross-assay objective moves unseen-pair CRPS by −0.025 (bar 0.067) and denoising by +0.007 (bar 0.103): neither is statistically significant (Fig. 4a). Correction alone scores 0.667 (0.612 + 0.055, seed range 0.019); cross-assay 0.642 (0.603 + 0.039, 0.034). At weight 4, cross-assay nearly clears (−0.034 against 0.039), mostly in scale error (−0.029). The scrambled twins are worse on unseen pairs by +0.322 and +0.282, but the planned twin check on trained pairs cannot fail, because the twin is scored on its own training labels. Reading: against the claim, weak evidence.

Pivot through the base: no general gain.

Routing beats the direct route beyond the bar in 2 of 98 unseen cells (66 needed; 2 of 90 cross-setting cells) and loses in 0. Pooled, it is lower in 6 of 6 seeds (gain +0.027 against bar 0.154): direct 0.660 (0.610 + 0.050, seed range 0.031), chained 0.633 (0.576 + 0.057, 0.077; Fig. 4b). The chained route follows the depth law (0.089, seed range 0.140; bar 0.16). On six seeds the trained-pair control missed its band (−0.006 against 0.006); the PI counted it on the three seeds shared with the reference model. Unplanned, the agreement-loss model's direct route scores 0.598 (0.577 + 0.021, 0.015), +0.062 against bar 0.061; this is exploratory. Reading: against, moderate evidence.

Mappability input: a local gain.

In the lowest-mappability tenth of bins the mappability channel lowers CRPS on all 6 of 6 tracks for read length 36 (0.303, 0.300 + 0.003, seed range 0.010, against 0.353, 0.322 + 0.031, 0.004) and for mapping filter 0 (0.532 against 0.590; Fig. 4c). It beats shifted mappability (+0.062 against bar 0.005), the gain is local (genome-wide 0.665 against 0.660), and a DNA-sequence input gives about a quarter of it. The depth arm, meant to stay unchanged, improved by −0.0094 (seed standard deviation 0.0011), so this no-change control failed. Reading: supports, weak evidence.

4 Conclusion

On trained pairs the covariates carry the information: for depth they close most of the gap to the oracle, and for run type they beat a covariate-free map. On unseen pairs the gain shrinks and no direct map follows the depth law. Neither a joint objective nor routing through a shared base changed this; a mappability input helps where reads map poorly. These are readings: a pre-registered pass of the transfer test on more tracks would settle the question (Table 1).

Limitations.

Only 7 tracks and 3 to 6 seeds; the final checks had a joint probability of 41% to 51% of passing if true; the twin check of the joint objective could not fail; the depth oracle was redefined, and the run-type bar set, after their runs; run type and control settings have no valid ceiling.

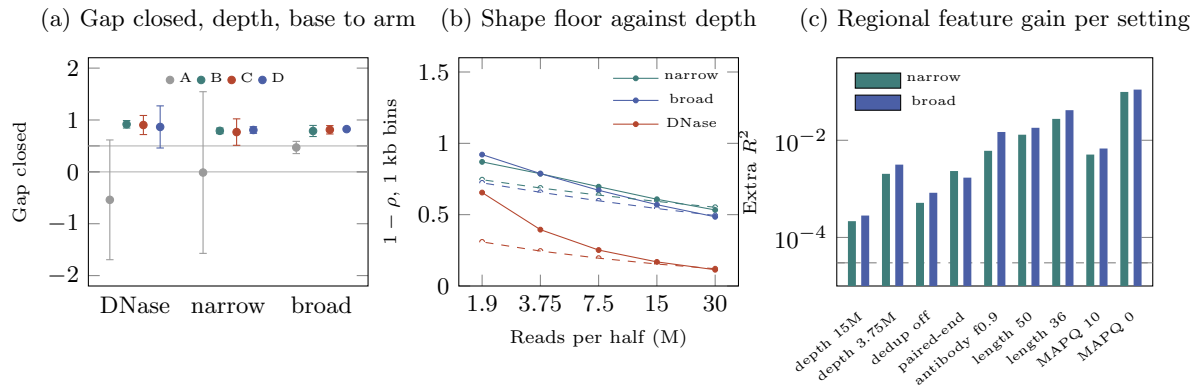


Fig. 2 What the settings do, and what trained pairs reach. (a) Gap closed from one-warp to the oracle on depth pairs, base to arm, one generator per track; error bars are 2× the seed range in gap units; grid lines mark zero and the 0.5 bar. (b) The 1 kb shape floor (median over products) along the thinning ladder; solid lines thin whole reads, dashed lines thin single counts; the rightmost point is extrapolated; the seed range (at most 0.003) is smaller than the marks. (c) Extra R^2 that GC, mappability, repeats and peak context add to the signal-level null (median over tracks); the dashed line is the largest shuffle control.

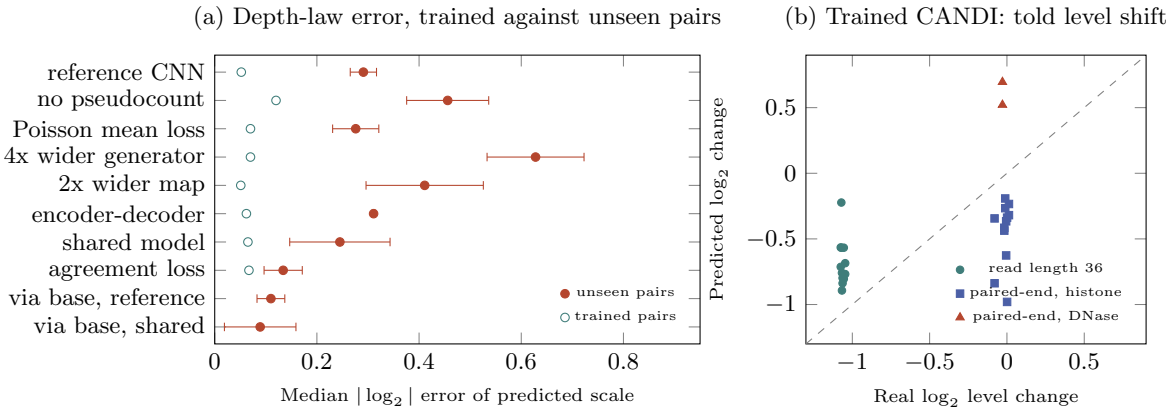


Fig. 3 The correction does not transfer. (a) Depth-law error of each fix tried (one generator per track except the shared models), on trained pairs (open) and unseen arm-to-arm pairs (filled; error bars span the seed range). The two bottom rows chain trained maps through the base. (b) Predicted against real chromosome-wide level change when a trained CANDI is told an arm's covariate, one point per track and seed; the dashed line is agreement.

Next steps for the mini-CANDI prototype, without ranking.

The design rulings keep count space only and depth as a learned probe, and defer costly reprocessing.

- **A first prototype** with depth and a mappability channel, gated by the twin, the seed range and the unseen-pair depth law. Testbed scale (training 11 to 36 min per arm); teaches whether correction survives inside an imputer.
- **A pre-registered test of the agreement loss.** One arm, 6 seeds; teaches whether the exploratory lead is real.
- **Run-type size on more tracks**, from single-end reprocessed challenge tracks. CPU analysis, if the PI opens that panel; teaches how large the challenge's failure knob is per assay.
- **New counterfactual pairs for CANDI's corpus.** Read-level settings need the pipeline rerun (median 13.4 h per product); teaches whether real pairs fix transfer.

References

[1] Schreiber, J.M., Boix, C.A., Lee, J.W., et al.: The ENCODE Imputation Challenge: a critical assessment of methods for cross-cell type

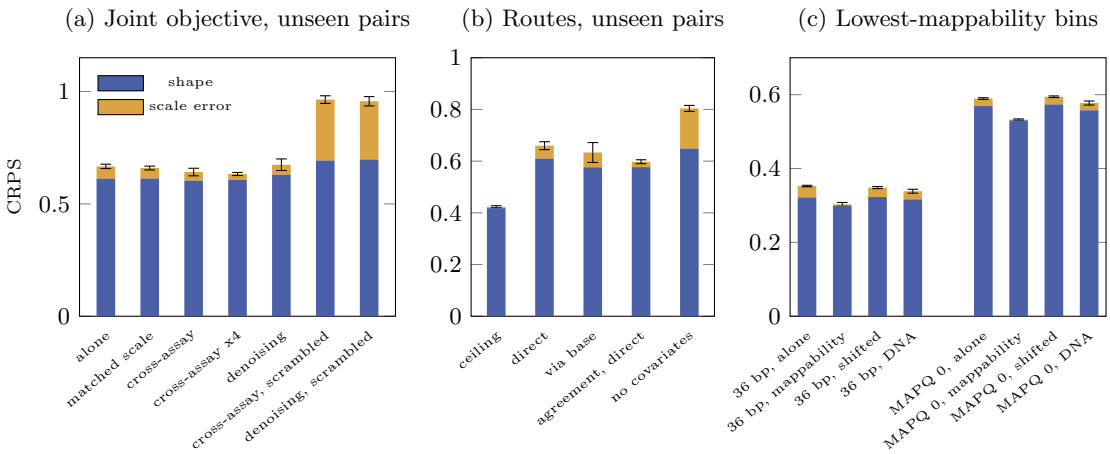


Fig. 4 The final experiments on one shared model. Each bar is CRPS split into shape (blue) and scale error (gold); error bars span the seed range over 3 seeds (6 for the routes). (a) Correction alone, a matched-loss-scale control, the two joint objectives at weight 1 (cross-assay also at 4) and their scrambled twins. (b) The base-to-target ceiling, the direct and chained routes, the agreement-loss model and the no-covariates floor. (c) Mean over tracks in the lowest-mappability tenth of bins.

Table 1 Each goal, its status and one line of evidence. Met: the question has a full reading. Partly met: part of it is read. Answered: the aim was not met, but the reading answers the question well enough to act. Every reading is exploratory.

Goal	Status	Evidence
Do covariates map one measurement onto another? (main question)	Partly met	Beat the scrambled twin on trained pairs; count gain 0.29 falls to 0.11 unseen; swap ($\leq$ 0.22 against bar 0.1) and depth-law checks unmet
Depth conditioning beyond one-warp	Met	Designs B–D close 0.77–0.92 of the gap; gain above 2× seed range in 18/18 cells; design A no gain
Run-type conditioning beyond one-warp	Partly met	Beats one-warp in 16/16 histone cells by 0.055–0.213; no valid ceiling; DNase 1/8
Read-selection settings	Partly met	Count gain over twin 0.08–0.18 trained (seed range 0.003–0.059); one-warp bar not scored
Control and caller settings	Set aside	p gain 0.36–0.62 trained, 0.31–0.48 unseen (seed range $\leq$ 0.024); out of scope once the work moved to count space
Capacity of the map	Answered	A monotone curve suffices in counts; more capacity does not fix unseen pairs
Depth offset	Answered	Helps design A only: DNase 3.06 $\pm$ 0.113 (1.36 + 1.70) to 1.247 $\pm$ 0.016 (1.109 + 0.138), bar 0.227; other designs 0/36; depth law met in 0/64 cells
Counts-to-p map	Answered	Formula plus learned correction, log CRPS 0.0659 $\pm$ 0.0035 (0.0656 + 0.0003), against 0.1791 $\pm$ 0.0010 (0.1367 + 0.0423) affine; one track
Why the depth law is missed	Answered	Not pseudocount, loss, width or encoder-decoder (0.24–0.63 unseen); chaining meets it (0.09–0.11); arm-to-arm training fixes scale, not CRPS (-0.53)
Does trained CANDI correct other covariates?	Answered	Level moves when told read length or run type, not as the real tracks do; per-region tracking unreadable (depth control failed)
Which settings move a track most	Met	Peak height above floor 27/27 (33–763×), shape 3/27
How much resampling alone moves a track	Met	1 kb shape floor at 30M: narrow 0.533, broad 0.485, DNase 0.114 (seed range $\leq$ 0.003); level, not region, sets it
Where along the genome settings act	Met	Depth uniform; mapping filter 0.099–0.110, read length 0.013–0.042 extra R^2, in low-mappability bins
Joint objective changes transfer	Not supported	Cross-assay -0.025 (bar 0.067), denoising $+0.007$ (bar 0.103); weak
Routing through the base transfers	Not supported	2/98 cells; pooled gain $+0.027$ against 0.154; moderate
Mappability input helps	Supported	6/6 and 6/6 tracks in the lowest tenth; depth control failed; weak

imputation of epigenomic profiles (2023)